DATA ANALYSIS SKILLS

What Makes a Film Highly Rated on IMDb?

Binary Logistic Regression Analysis of IMDb Film Ratings

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Introduction

Background

IMDb ratings shape millions of viewing decisions each year.

A rating above 7 out of 10 is widely recognised as a quality threshold — used by critics, streaming platforms, and audiences alike.

Understanding what predicts this threshold can reveal how film properties relate to audience reception.

Research Question

Which film properties predict a rating above 7 on IMDb?

Variables of Interest

`year`	Release year
`length`	Film duration
`budget`	Production budget
`votes`	Number of viewer votes
`genre`	Film genre (7 categories)

Data Overview & Cleaning

1,937
Films in original dataset

1,834
Films after removing missing values

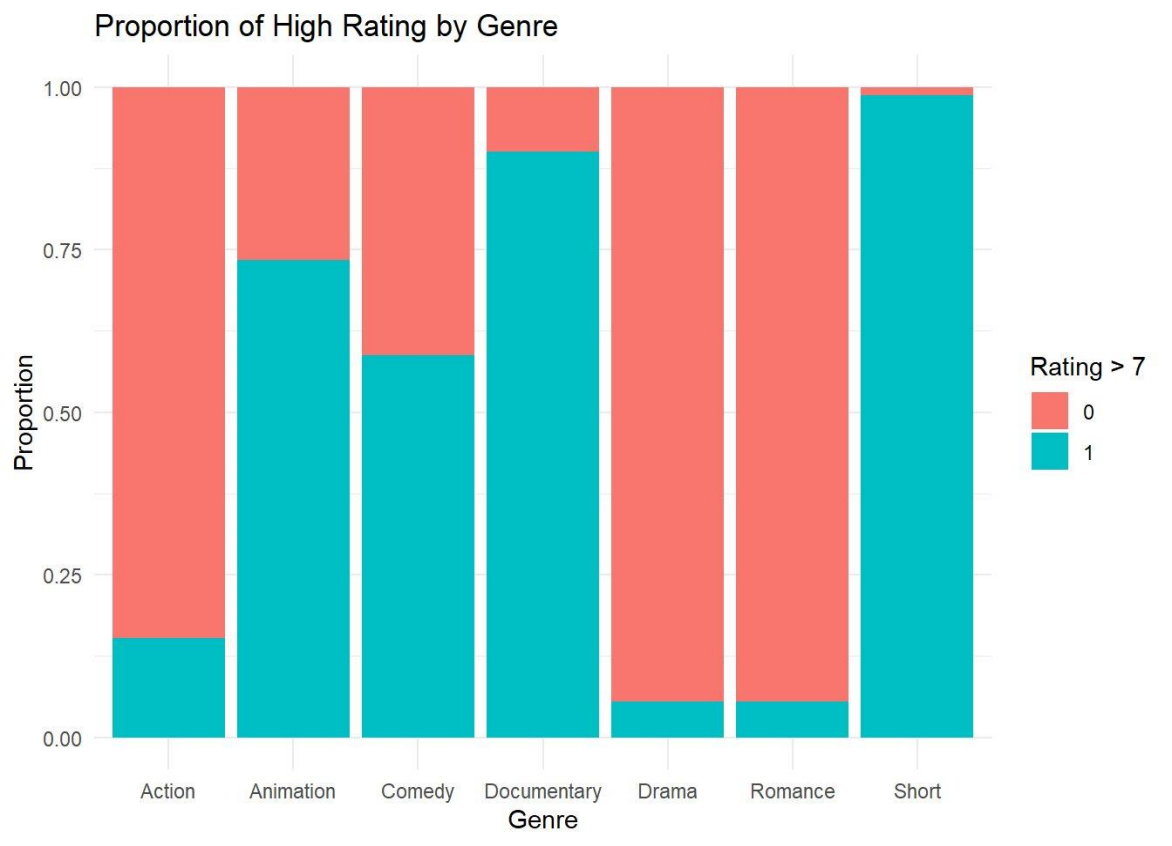
606
High-rated films (33%)

1,228
Low-rated films (67%)

Data Preparation Steps

①	**Missing values removed**	103 films had missing length values → removed using na.omit()
②	**Binary response created**	high_rating = 1 if rating > 7, else 0
③	**Log transformation**	votes was heavily right-skewed → log(votes) used to reduce skewness

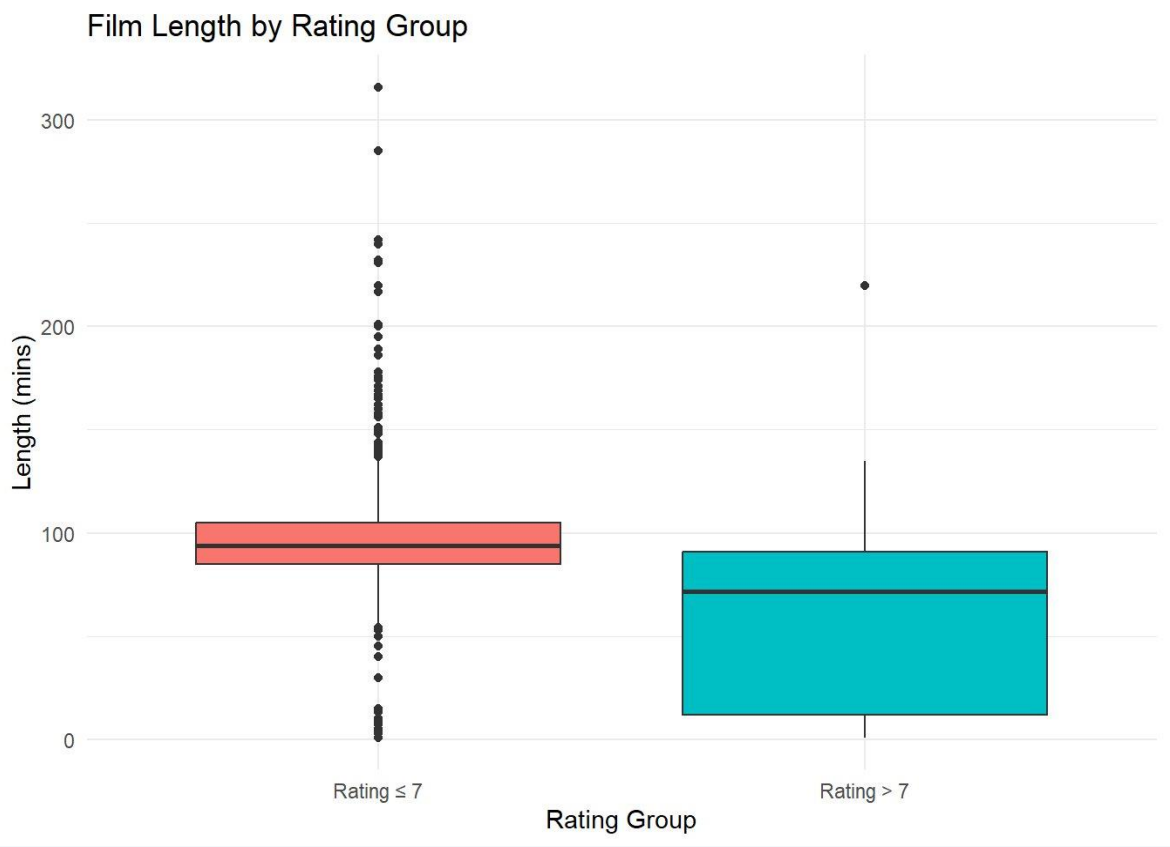
EDA: Proportion of High Rating by Genre



Key Takeaway

- Short & Documentary have the highest proportion of high-rated films (~90%+)
- Drama & Romance have the lowest — below 10%
- Genre shows the clearest separation between rating groups

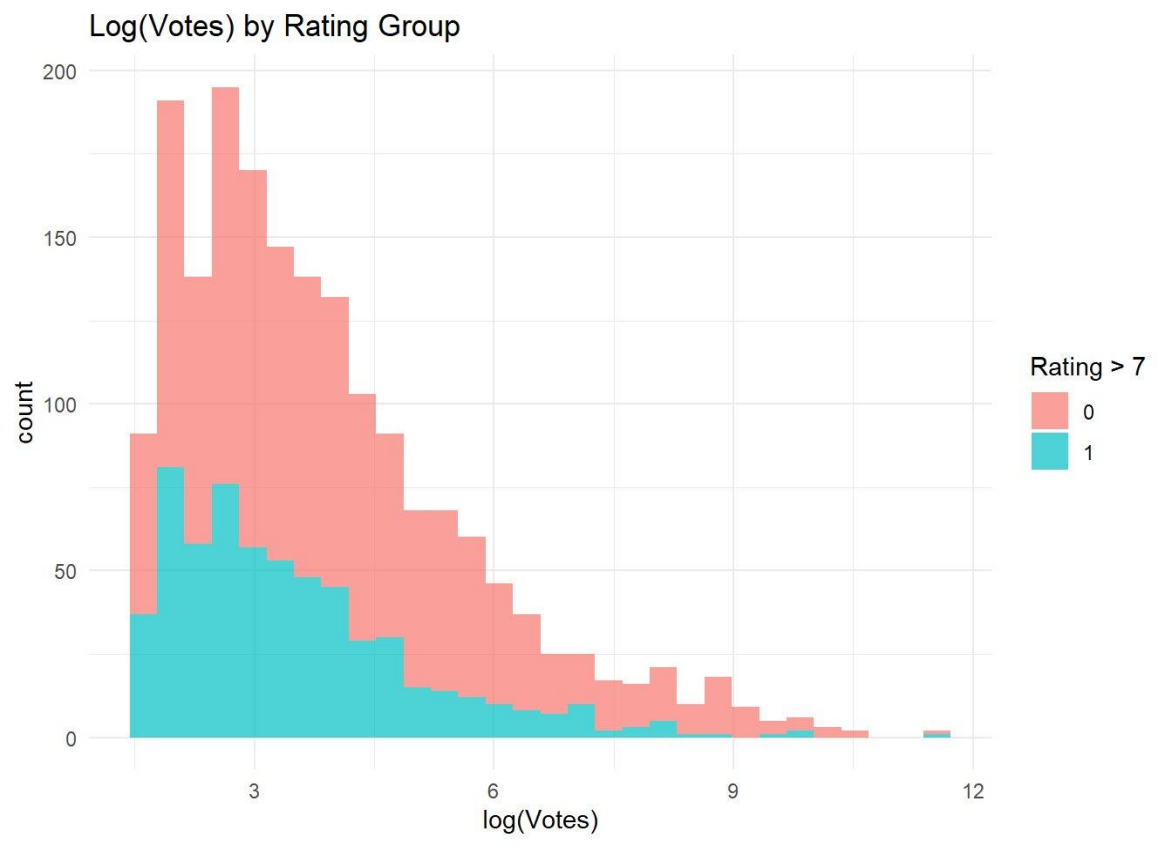
EDA: Film Length by Rating Group



Key Takeaway

- High-rated films tend to have a lower median length
- High-rated group shows a wider spread (larger IQR)
- Suggests longer films are less likely to receive high ratings

EDA: Log(Votes) by Rating Group



Key Takeaway

- Distributions overlap substantially across both groups
- Both groups concentrated in the lower vote range
- Vote count alone does not discriminate between rating groups

Formal Analysis: GLM

Model Overview

Response Variable: high_rating (1 = rating > 7, 0 = rating ≤ 7)

Model Type: Logistic Regression (GLM with binomial family & logit link)

Purpose: To examine how film properties influence the probability of receiving a high rating

Formula

Model 1

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 year_i + \beta_2 length_i + \beta_3 log_votes_i + \beta_4 budget_i$$

Model 2

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{year}_i + \beta_2 \text{length}_i + \beta_3 \text{log_votes}_i + \beta_4 \text{budget}_i + \sum_j \gamma_j \text{genre}_{ij}$$

Formal Analysis: GLM

Model Comparison Using AIC

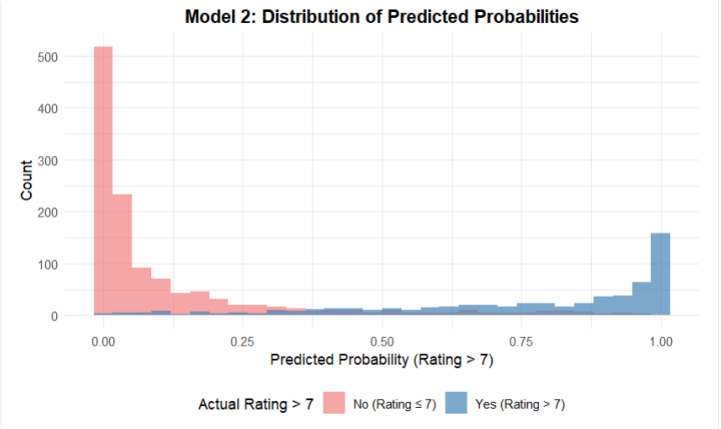
Model 2 has a much lower AIC, indicating a better fit

Genre contributes significantly to explaining high ratings

	df	AIC
model1	5	1594.8069
model2	11	969.9133

Deviance Analysis

The null deviance for Model 2 was 2327.27, while the residual deviance was reduced to 947.91



Model 2 also has strong discriminative power

Formal Analysis: GLM

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-22.7749704	7.5423683	-3.0196046	0.0025310
year	0.0100553	0.0038376	2.6201897	0.0087881
length	-0.0650261	0.0048197	-13.4917972	0.0000000
log_votes	0.0097877	0.0519654	0.1883511	0.8506014
budget	0.5031948	0.0368151	13.6681601	0.0000000
genreAnimation	-0.4890389	0.4347951	-1.1247573	0.2606919
genreComedy	3.1193441	0.2236778	13.9457054	0.0000000
genreDocumentary	5.2430893	0.4951489	10.5889142	0.0000000
genreDrama	-1.6423418	0.2921991	-5.6206251	0.0000000
genreRomance	-1.9437411	1.8077313	-1.0752378	0.2822683
genreShort	3.5013784	1.0804758	3.2405894	0.0011928

Key Findings from Final Model

Significant Predictors:
Year(positive),Length(negative),Budget (positive)

Genre Effects :
Higher odds: Comedy, Documentary, Short
Lower odds: Drama

No significant difference: Animation, Romance

Results

key takeaway

The GLM results show that genre and budget are the strongest predictors of high ratings

Higher budget films are significantly more likely to receive ratings above 7

Film length has a negative effect, ms are less likely to be highly rated

Year has a small positive effect, indicating newer films are slightly more likely to achieve high ratings

The number of votes (log_votes) is not statistically significant and does not strongly influence rating outcomes

Conclusions & Future Work

Summary of Findings and Implications

Film genre plays the most important role in predicting whether a film receives a high IMDb rating

Budget is also a strong positive factor, while film length shows a negative relationship with high ratings

Overall, the model suggests that content type and production scale are more important than popularity (votes)

Limitations and Future Work

The dataset does not include other important factors such as actors, directors, or audience reviews

Future analysis could include additional variables such as cast, director, or user reviews ,And More advanced models (e.g. random forest) could be used to improve prediction accuracy